MIL-T-0052864C(84)
20 January 1988
USED IN LIEU OF
MIL-T-52864B
22 APRIL 1987

MILITARY SPECITICATION

TRUCKS, LIFT, FORK: DIESEL-ENGINE-DRIVEN, PNEUMATIC-TIRED, 10,000 CAPACITY AT 48-INCH LOAD CENTER

"This military specification has been prepared by WR-ALC/MMVRG based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of MIL-T-52864. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition."

1. SCOPE

1.1 Scope. This specification covers the general requirements for diesel engine-driven forklift trucks.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specification and standards of the issue listed in that issue of The Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMVRG, Robins AFB, GA 31098-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3930

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SPECIFICATIONS

FEDERAL

W-B-131	Battery Storage (Vehicular, Ignition, Lighting and Starting)

MILITARY

MIL-V-173	Vehicle, Moisture And Fungus Resistant (For Treatment of Communications, Electornic And Associated Equipment
MIL-P-514	Plates, Identification, Instruction, and Marking, Blank.
MIL-T-5624	Turbine Fuel, Grades JP-4 and JP-5
MIL-A-11755	Antifreeze, Arctic Type

MILITARY

MIL-A-8421	Air Transportability Requirements, General
MIL-P-23377	Primer Coating, Epoxy-Ployamide, Chemical and Solvent Resistant
MIL-A-46153	Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Single Package
MIL-B-46176	Brake, Fluid, Silicone, Automotive, All Weather, Operational and Preservative
MIL-P-52192	Primer Coating, Epoxy
MIL-E-52649	Engine Cold Starting Aids, Ether Fuel Primers
MIL-E-52798	Enamel, Alkyd Camouflage
MIL-P-53022	Primer, Epoxy Coating, Corrosion Inhibiting, Lead and Chromate Free
MIL-P-53030	Primer Coating, Epoxy, Water Reducible, Lead and Chromate Free
MIL-T-83133	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8
MIL-H-83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base
MIL-C-83286	Coating, Urethane, Aliphatic, Isocyanate, for Aerospace Applications

STANDARDS

FEDERAL

W-B-131	Battery Storage (Vehicular, Ignition, Lighting and Starting)

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MILITARY

MIL-STD-130	Identification Marking of US Military Property
MIL-STD-209	Slinging and Tiedown Provision for Lifting and Tying Down Military Equipment
MIL-STD-248	Welding and Brazing Procedure and Performance Qualification
MIL-STD-461	Electromagnetic Emission and susceptability Requirements for the Control of Electromagnetic Interference
MIL-STD-1400	Engine, Gasoline or Diesel, Methods of Test
MIL-STD-1474	Noise Limits for Army Material
MS51336	Lunette - Coupler Drawbar, Ring

2.1.2 Other Government documents, drawings, and publications. The following other Government document forms a part of this specification to the extent specified herein.

DEPARTMENT OF LABOR (DOL)

29DF1910 - Occupational Safety and Health Administration

(Application for copies should be addressed to a local DOL officer or the Superintendent of Document, US Government Printing Office, Washington DC 20402.

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

B56.1	- Safety Standard for Powered Industrial Trucks
MH11.3	- Load Handling Symbols for Powered Industrial Trucks.
Z26.1	- Safety Glazing Materials for Glazing Motor Vehicles Operating on Land Highways

(Application for copies should be addressed to the American National Standards Institute Inc., 1430 Broadway, New York NY 10018).

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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086).

UNDERWRITERS LABORATORIES INC. (UL)

UL 558 Standard for Industrial Trucks, Internal Combustion Engine-Powered.

(Application for copies should be addressed to Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

DESIGN HANDBOOK

AFSC DH 1-11 Air Transportability

(Copies of this design handbook should be obtained from HQ AFLC, ASD/ENECA, Wright-Patterson AFB OH 45433-5000).

2.3 Order of precedence. In the event of a conflict between the text of the specification and the reference cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The trucks shall be diesel-engine-driven, front wheel drive, rear wheel steer, DS rated per UL 588, fork-lift trucks.

3.2 First article. The contractor shall furnish two complete trucks; one for first article evaluation and demonstration in accordance with 4.3 and 4.4 and the other for user testing as defined in 4.8, and within the timeframe specified (see 6.2). All testing shall be observed and approved by Government representatives. Both approved first article trucks shall be rehabilitated, to a new condition and delivered as specified in the contract. The trucks furnished for both first article evaluation and user testing shall have Type A winterization.

3.3 Environmental requirements.

3.3.1 Operating temperature. The truck shall start without preheating within five minutes in any ambient temperature (and

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with the truck stabilized at ambient temperature) from -25 degrees F to +125 degrees F. With proper lubricants, the truck shall operate as specified herein within 15 minutes after engine start in any ambient temperature from -40 to +125 degrees F.

3.4 Safety. The forklift shall conform to the requirements of ANSI B56.1, UL 558 Type DS and OSHA Standards in effect at the time of manufacture. The truck shall be tested in accordance with UL 558 to prove that it meets the DS requirements. The results of this testing shall be included in the First Article Test Report. Upon successful completion of these tests, all production forklifts shall be marked with corrosion resistant plates as required by UL 558 for DC trucks.

3.4.1 Nuclear safety. When nuclear safety is specified (see 6.2), the trucks shall be equipped with additional safety features as specified herein.

3.4.2 Fire extinguisher. When nuclear safety is specified (see 6.2), a fire extinguisher shall be mounted on each truck in an easily accessible location. The fire extinguisher shall be a 2 1/2 pound Halon 1211, painted the same color as the forklift exterior.

3.4.3 Rearview mirror. When specified (see 6.2), trucks shall be equipped with a rearview mirror (with mounting arm) mounted on the left-hand side within the operator's field of view as described in SAE J985.

3.4.4 Seat belt and operator's restraint. An operator's seat belt conforming to SAE J386, type one, shall be installed. Additional restraint devices (e.g., winged seats, enclosures, etc.) designed to ensure the operator's body remains entirely within the protection of the frame and overhead guard in the event of tipover or lateral roll will also be provided. The restraint system will not interfere with operator access or mobility or the vehicles operation. A warning decal will be provided advising of the hazards of tipover and the importance of using the restraint system. The installation of the lap belts alone will not be considered as adequate occupant restraint. The manufacturer will provide evidence of the suitability and effectiveness of any provided occupant restraint systems.

3.5 Human factors engineering. The characteristics of the truck shall provide for operation by the 5th to 95th percentile male and female in accordance with SAE J833a.

3.5.1 Maintenance operation. Unless otherwise specified herein, each operation in the following list shall be accomplished by one man in not more than the time specified, using common tools and special tools, if any, furnished with the truck:

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a. Remove, replace and adjust all engine driven belts - 3/4 hour.

b. Remove and replace alternator - 1/2 hour.

c. Remove and replace all filters, screens and strainers in all hydraulic systems - one hour.

d. Remove and replace engine coolant system hoses - 1/2 hour.

e. Drain engine lubricating oil, remove and replace oil filter elements and refill crankcase - one hour.

f. Remove and replace all fuel filter elements - 1/2 hour.

g. Disconnect battery cables, remove and replace batteries and reconnect battery cables - 1/2 hour.

h. Drain torque convertor oil and transmission oil, remove and replace all filter elements and strainers and refill convertor and transmission - one hour.

i. Remove and replace starter - one hour.

j. Bleed and adjust brakes and refill master cylinder (if applicable) - two men - 3.4. hour.

3.5.2 Special tools. Any special tools (tools that are peculiar to this truck) that are needed to perform scheduled or unscheduled maintenance shall be provided by the contractor. One set of special tools shall be supplied with each truck.

3.6 Engine. The diesel engine shall be a commercial type. The engine shall operate on diesel fuel, JP-4 and JP-5 conforming to MIL-T-5624 and JP-8 conforming to MIL-T-83133. Engine manufacturer's certification is acceptable. Emergency manual shutdown provisions shall be furnished when recommended by the engine manufacturer. Exhaust smoke shall not exceed 3.0 using a Robert Bosch Smokemeter when tested in accordance with Test Method 2300 of MIL-STD-1400. The contractor of the engine manufacturer shall perform this test and include the results in the First Article Test Report. The total power consumed by all engine driven components at their maximum output shall be not more than 80 percent of the engines net power output rating at 2,000 feet elevation determined in accordance with SAE J1349. The engine manufacturer's certified net brake horsepower and

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torque versus engine RPM shall be included in the First Article Test Report along with the maximum total power required to operate the truck and how it was calculated.

3.6.1 Engine accessories.

3.6.1.1 Cold starting assist. Glow plugs or a measure shot ether injection system shall be furnished to assist in meeting the cold starting requirements of 3.3.1.

3.6.1.2 Alternator. The alternator shall provide at least five amperes to charge the battery at engine idle (as specified by the engine manufacturer) and with the full electrical load, except the starting system, activated. The alternator shall be DS rated ANSI/UL 558.

3.6.1.3 Starting system. A device shall be furnished to prevent reenergizing the starter motor after the engine has been started. Trucks shall be provided with an interlock in the starting system or other means to prevent energizing the starter motor, except when the transmissions in the neutral position. The starter shall be heavy duty and shall be in accordance with the engine manufacturer's recommendations.

3.6.1.4 Fuel system. The capacity of the fuel tank shall provide for not less than 8 hours operation when demonstrated as specified herein. The tank shall be equipped with a safety filler-cap assembly painted with green enamel which shall incorporate a self-closing cap, screw-locking device for the cap, removable strainer, and shall provide for padlocking. The fuel tank filler neck opening shall have an inside diameter of at least 3.0 inches. A data plate in accordance with 3.24 or truck marking in accordance with 3.25 shall be furnished and marked "Diesel Fuel" in not less than 1-inch letter height located near the fill pipe. The fuel tank shall be equipped with a hand operated drain cock at the lowest point of the tank. The drain cock shall be protected by a guard, or be so located to prevent damage. The drain cock shall be located, or a drain line provided, so that a five gallon container on the ground can take the drained fuel.

3.6.1.5 Air induction system. A heavy-duty, dry-type replaceable air cleaner shall be furnished. The air cleaner shall be constructed so as to permit disassembly without tools for servicing and shall be accessible when the engine cowl, hood, or access plate is opened. A restriction indicator shall be furnished.

3.6.1.6 Governor. A speed-limiting type governor shall be furnished. The engine speed shall not exceed the engine manufacturer's recommended intermittent duty speed.

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3.6.1.7 Servicing. The following items shall be easily accessible for inspection and servicing when the engine is at operating temperature without subjecting personnel to possible injury, burns, etc. They shall be in an open area, or covered by a hinged panel that can be opened or closed without the need of tools, by one man, within one minutes (for each panel):

a. Engine oil and dipstick and oil filler.

b. Transmission fluid dipstick and filler tube.

c. Radiator level indicator and filler.

d. Power steering pump fluid level indicator and filler.

e. Brake fluid reservoir.

f. Battery.

g. Hydraulic reservoir fluid level indicator and filler.

3.7 Transmission. The transmission shall be the full-torque, powershift type in combination with a torque converter. The transmission shall provide the gear ratios specified on the applicable specification sheet, all of which shall be manually selected. All transmission gears shall be constant mesh. Shifting between gear ratios shall be accomplished by selective engagement of multiple-disk clutches which shall be actuated by annular hydraulic pistons. Actuation shall be controlled by a hydraulic-control-valve body mounted on or in the transmission housing. The transmission shall provide for shifting under the full engine power through all successive gear ratios in each direction. the transmission shall provide controlled application of the clutches to provide smooth shifting without slippage that could damage the clutches. Oil pressure for actuation of the transmission clutches for torque converter supply, and for transmission lubrication shall be supplied by a pump mechanically driven by the engine. The transmission and torque converter fluid shall be filtered by full-flow filter(s), having replaceable elements. The transmission shall provide for positive inching control or delutch control of the truck (throughout the entire range of engine speed) in the forward and reverse directions. The inching or declutch control shall permit tilting and full lifting speeds with rated load while the transmission control is in forward or reverse position without vehicle movement when tested as specified herein.

3.7.1 Directional controls. Selective forward and reverse directional controls actuated by the operator's foot are not acceptable.

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3.8 Hydraulic system. The hydraulic system shall consist of all hydraulic components necessary for operation of the mast and forks and does not include the transmission. The system shall provide for lowering of rated load at rate specified in 3.22.2 in the event of failure of or damage to hydraulic hose supplying the lift cylinder. A pipe tap or straight-thread O-ring boss test point shall be located at the outlet of each pump or pressure inlet of each control valve for measuring pressure. When nuclear safety is specified (see 6.2), the hydraulic lift system shall be fail-safe (see 6.4). The hydraulic fluid shall conform to MIL-H-83133. The hydraulic system shall be compatible with this type fluid. The hydraulic reservoir shall have a filler opening of at least 2.0 inches inside diameter. There shall be an hand operated drain cock located at the lowest point of the reservoir. The drain cock shall be protected by a guard, or be so located to prevent damage. The drain cock shall be located, or a drain line provided, so that a five gallon container on the ground can take the drained fluid.

3.8.1 System filtration. System filter(s) that meet the performance requirements of MIL-F-52723 shall be furnished in the suction line, the return line or both.

3.8.2 Relief valve. A hydraulic-system relief valve shall be provided to protect the hydraulic system from excessive pressure.

3.8.3 Hydraulic control valves. Mounting of the valves shall be rigid and shall permit no vibration that may cause instability of the controls. All control-valve spools shall be spring centered with sufficient force to return the levers to the center position and retain them there throughout all operations. Spools shall not bind under any operating conditions specified herein. All exposed control valve surfaces shall show no evidence of corrosion when demonstrated as specified herein.

3.8.4 Hose reels. When hydraulic hose reels are used on the mast to maintain hose tension during mast extension, sheave wheels, guide rollers, guide blocks, or other means shall be provided to prevent the hose from becoming entangled around and riding over the reels or reel flanges. Hose reels shall be mounted such that they are within the plan outline of the truck for all positions of the forks.

3.8.5 Suction line. The suction line shall be sized so that the minimum pump inlet pressure does not fall below atmosphere by more than five inches of mercury at 140 to 160 degrees F oil temperature and at maximum governed engine speed.

3.8.6 Hydraulic pump. The hydraulic pump(s) shall

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withstand a minimum proof purchase of 1 1/2 times the maximum system working pressure without permanent deformation, damage or external leakage.

3.8.7 Filter(s). The filter housing shall have a minimum proof pressure of 100 psi for a return line filter, or 30 psi for a suction line filter, without permanent deformation, damage or leakage. At maximum system oil flow through the filter, including lowering rated load on the forks at maximum governed engine speed and an oil temperature of 125 to 155 degrees F, the pressure drop across a new return line filter assembly shall not exceed 10 psi or 0.75 psi for a suction line type filter. An automatic bypass shall be provided in the filter and shall open at not less than 10 psi greater than the pressure drop measured across a new filter assembly for a return line filter or not less than 2.0 psi for a suction filter. The bypass shall also permit full flow at a pressure not greater than 35 psi for a return line filter or not more than 3.5 psi for a suction filter. The filter shall be equipped with a contamination indicator that shall show when the element is clean and when the filter is beginning to bypass, requiring a change of the element. The indicator shall continue to show a need for element change after the hydraulic system pressure is reduced to zero. The indicator shall be clearly readable without the removal of any components.

3.8.8 Cylinders. All cylinders shall withstand a pressure of 1.5 times the system relief valve setting without evidence of rupture, permanent deformation, damage or external leakage.

3.8.9 Contamination levels. When hydraulic system is tested as specified in 4.4.28, particle counts shall not exceed 1,000 particles per milliliter greater than 10 micrometers, and particle counts shall not exceed 10 particles per milliliter greater than 20 micrometers.

3.8.10 Pressure hose assemblies. Permanently attached type pressure hose couplings are acceptable as original equipment hose assemblies. The truck shall be compatible with hose assemblies with field attachable reusable fittings in place of each hose assembly with permanently attached couplings when tested as specified herein.

3.9 Uprights and carriage assembly. Uprights shall be of the telescopic, roller type. The drive tires shall not interfere with the carriage assembly travel at any upright tilt angle. Chains shall be installed in such a manner that they cannot be displaced from the roller until they have been raised a minimum of 1/2 the chain side plate height. Crossmembers of the mast structure, when in a lowered position, shall not interfere with the operator's horizontal line of sight, plus or minus 10 degrees, to the front of the truck. Without leaving his seat,

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any lift height and with the tines at any fork spacing specified by item 6 of MIL-T-52864/6A, when there is no load on the forks. Means shall be provided in the uprights to compensate by replacement or adjustment of wear which may develop between lateral rolling contacts. Positive means shall be provided to prevent overtravel of the carriage or channel in both upper and lower positions. Carriages shall be supplied with adjustable thrust bearings to resist thrust introduced by off center loading. Chains used in the lifting mechanism shall be a listed ANSI Standard and shall have a factor of safety of not less than 4.6 to 1 based on average ultimate strength. When nuclear safety is specified (see 6.2), load securing rings or eyes shall be provided on both left and right sides of carriage and located not more than 12 inches from top of carriage assembly. Safety factor for load securing devices shall be not less than 3 to 1, based on yield strength, assuming that each device is loaded equally. Ring or eye openings shall be determined using MIL-STD-209. The load securing device, when not in use, shall not extend forward of any surface on front face of carriage. The carri- age. The carriage assembly/load securing device combi- nation shall be capable of withstanding a horizontal force (at load securing devices) equal to three times the truck capacity without failure or permanent deformation of carriage assembly. Direction of force is forward. The carriage or load backrest shall prevent small containers, approximately three inches on a side, from contacting the hydraulic components of the carriages or mast. The mast shall have two lift cylinders, each located near or within the mast channels, instead of one lift cylinder located in the middle of the mast. The reason for this is to allow an open area in the mast for increased operator's visibility. The open area between the structure of the mast (measured between the channels, lift cylinders, or whatever is closest together) shall be at least 30 inches wide. It is allowable to have hydraulic hoses, and not more than two chains, within this open area, but they shall be located as far as possible away from the center of the mast.

3.9.1 Forks. Forks shall conform to ANSI B56.1 and the applicable specification sheet. The fork spacing detailed on the specification sheets shall be measured from the outside to out- side surfaces of the forks. Fork thickness shall not exceed 1/2 inch at the tips and the end taper shall be not less than 14 inches long. Taper shall be on the underside of the forks. The top of the fork tips shall be chamfered or rounded to break corners and provide smooth edges.

3.9.2 Fork positioner. Hydraulic powered fork positioners shall be furnished. Fork spacing shall be simultaneously, in synchronization, hydraulically adjustable within the dimension specified on the applicable specification sheet. The dimensions shall be measured from the outer edges of the forks.

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3.9.3 Side shift. Trucks shall be equipped with a hydraulically powered side-shift mechanism. The side-shift mechanism shall side shift rated load a minimum of 6 inches each side of center (12 inches total travel). The forks shall move in a synchronization when side-shifted. When side-shifting or moving for lateral spacing (see 3.9.2), the fork tips shall remain at the same horizontal height over the entire range of movement. The surfaces that allow lateral movement of the carriage and forks shall be capable of accepting lubrication with no disassembly.

3.10 Steering. An automotive type steering wheel not to exceed 18 inches in diameter, mounted on the steering column, shall turn the steering wheels. There shall be a minimum of 7 inches measured vertically between the lowest point on the steering wheel rim and a line parallel to the floor and passing through the highest point of the unoccupied seat cushion. There shall be a minimum of 3 inches between the rim of the steering wheel and the directional control lever(s) for all positions of the lever(s). Wheels shall steer true in accordance with the Ackermann steering principle. Clockwise rotation of the steering wheel shall provide for right-hand turning of the truck while in forward motion. Provision shall be made for precision adjustment for wear and alignment of all major steering components and for lubrication of all friction points by accessible grease fittings or permanent lubrication. The power steering reservoir shall be accessible for filling and checking of the lubricant level. The truck shall negotiate the obstacle course in the reliability test with only the displaced wheel off the ground at any time. Means shall be provided to prevent wheels or tires or steering linkage from contacting any part of truck structure, other than stops, irrespective of obstacle magnitude or cramped steer angle. The steering wheels shall be capable of being turned from extreme right to extreme left or vice versa in not more than six complete turns of the hand steering wheel. Power steering shall be furnished. With no load on the forks, engine at idle speed, service brake fully applied and truck not moving, the tangential force on the steering wheel required to cause the truck steering wheels to turn shall not exceed 10 pounds.

3.11 Brakes.

3.11.1 Service brake system. The brake system shall conform to ANSI B56.1. If hydraulic brakes/clutches are utilized, the brake system shall be filled with a fluid that is listed on the qualified products list of MIL-B 46176. Automatic brake adjustment shall be provided. Automatic shall maintain correct lining to drum or disc adjustment without looseness or lockup. The automatic adjusters shall be of the guarded or

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enclosed type, protected from environmental damage or malfunction.

3.11.1.1 Power brake system. The power source shall charge the system to maximum power setting within 5 seconds when the brake actuating energy is fully exhausted. An audible and visual low power warning device shall be furnished. The brake system shall be straight air (3.11.1.1.1), air-over-hydraulic (3.11.1.1.2), or hydraulic-over-hydraulic that uses the main hydraulic system, or a secondary hydraulic system, to supply the power assist.

3.11.1.1.1 Straight airbrake system. When a straight airbrake system is furnished, the system shall provide for the removal of water from the service air supply.

3.11.1.1.2 Air-over-hydraulic brake system. When air-over-hydraulic brake system is furnished, the complete air system shall be in accordance with 3.11.1.1.1.

3.11.1.2 Tires. The tires shall be pneumatic, high pressure, tube or tubeless, with all six tires identical. Drive wheels shall be duals and steering wheels singles. Drive wheel clearance to any part of the truck structure shall be sufficient to permit the use of standard tire chains. The tires shall be the type that will not pick up rocks, or other items less than 1.10 inch in diameter.

3.11.2 Parking and emergency stopping system. The parking, and when provided, the emergency stopping system shall be an isolated system actuating the service brakes in accordance with SAE J1152. The parking brake shall be hand operated in accordance with SAE J680. When nuclear safety is specified (see 6.2), The truck shall be equipped with an emergency braking system capable of stopping the loaded truck at maximum speeds when demonstrated in 10 accordance with 4.4.12. Stopping distance when using the emergency braking system only shall not exceed three times the stopping distance requirement for service brakes. If the parking brake is capable of meeting this requirement, a separate emergency brake is not required.

3.11.2.2 Parking brake. The parking brake shall activate a mechanical friction brake mounted on the drive or propeller shaft, or activate the service brake shoes of disk pads. the parking brake shall be hand operated in accordance with SAE J680. The parking brake shall be capable of holding the fully loaded truck, in both directions, on a 20 percent grade.

3.12 Truck access. Operator access to the truck shall be in accordance with minimum requirements of SAE J185.

3.13 Seat. A cushion seat and backrest shall be furnished

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in accordance with the minimum requirements specified in SAE J899. The seat location shall conform to SAE J898.

3.14 Cowl or hood. The engine shall be completely enclosed with a hood cover and side plates. Hood cover shall be capable of being opened or removed by one man using one hand. Handtools shall not be required for release of hood or side plates. The hood shall be securely held open by a positive lock- ing device.

3.15 Towing device. A pintle or pin-type towing device shall be provided in the center rear of the truck. The towing device shall accept lunette conforming to MS51336 without binding. The towing device shall not protrude beyond the path followed by the rear end of the truck when making a turn of minimum turning radius. The towing device shall be capable of withstanding, without physical failure, a towing force induced by the drawbar drag as specified in ANSI B56.1.

3.16 Batteries and battery cables.

3.16.1 Battery. Except when winterization is specified (see 6.2), the battery shall be the maintenance free type conforming to W-B-131. The battery shall be accessible and removable through a hinged panel that can be opened without the need for tools.

3.16.2 Battery cable terminals. Battery cables shall be in accordance with SAE J541. Positive and negative cable terminals shall be identified.

3.17 Instrument panel. The instrument panel shall be located so that the operator can read the instruments form his normal operating position. The instrument panel shall be in- stalled on an inclined or vertical plane to facilitate drainage of water. On this panel shall be mounted the ammeter, engine oil pressure gage, engine coolant temperature gage, transmission fluid temperature gage, fuel gage, engine run control, light switches, starter switch, airbrake pressure gage (if applicable), and any other instrument used. All gages, indicator lights, and switches shall be identified on the face of the gage or as specified in 3.24 as to nomenclature of component and fluid being monitored and positions and function control. International symbols are acceptable). All gages shall be mounted flush.

3.18 Controls and instrumentation. Each control shall be provided with a position marking diagram on a plate as specified in 3.24, or embossed or depressesd on or near the control. Marking for levers which control the lift mechanism shall con- form to ANSI MH11.3. The diagrams shall be visible to the operator from his normal operating position. All controls for the truck shall be located in the operator's compartment.

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Electric controls and instrumentation shall be moisture and weather resistant. Mechanical push-pull remote controls, when furnished, shall be of the waterproof anti-friction type, conforming to requirements for marine cables specified in SAE J917. The load motion controls (lift, tilt, etc.) shall be self cancelling (return to neutral when released). All instruments shall be internally lighted when the forward floodlights (see 3.19.1) are on and mounted on the instrument panel in view of the seated operator.

3.18.1 Light switches. A minimum of two light switches shall be furnished on the instrument panel. One switch shall control the forward floodlights) as specified in 3.19.1 and taillights as specified in 3.19.2. One switch shall control the rear floodlight(s) as specified in 3.19.1.

3.18.2 Engine run control. An engine run control shall be furnished and shall be a keyless type. It shall have not less than two positions: "on" and "off". The switch shall control the operation of monitors and of all electrical equipment except the horn. A mechanical engine shutoff may be furnished and shall be in addition to the emergency shutoff as specified.

3.18.3 Electrical system monitor. A remote indicating ammeter, shall be furnished. The ammeter shall be of sufficient capacity to indicate variation in current supply or demand experienced by the truck, exclusive of starter requirements.

3.18.4 Engine oil pressure monitor. A remote indicating engine oil pressure gage shall be furnished.

3.18.5 Engine coolant temperature monitor. A remote indicating engine coolant temperature gage shall be furnished.

3.18.6 Transmission temperature monitor. A remote indicating transmission temperature gage shall be furnished.

3.18.7 Hour meter. An electrically operated hour meter, which registers the number of engine operating hours only, shall be provided. The hour meter shall be capable of totaling 9999 engine hours. The hour meter shall be accessible for reading without removal of any component of the truck.

3.18.8 Fuel gage. An electric, thermally stabilized, remote indicating or mechanical fuel gage shall be provided.

3.18.9 Parking brake warning light. A red warning light shall be mounted on the instrument panel in view of the seated operator. It shall illuminate only when the parking brake is engaged and the engine run control (see 3.18.2) and battery disconnect switch are in the "ON" position.

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3.18.10 Battery disconnect switch. A manual disconnect switch readily accessible to the operator when in his normal operating position shall be installed in the battery and charging circuits to permit quick disconnect of the electrical system.

3.19 Electrical system. The truck shall have a nominal 12-volt electrical system. Multiple conductor routing shall be protected by flexible tubing or plastic electrical insulation tape. Grommets shall be provided whenever wiring passes through bulkheads, partitions or structural members; and the wiring shall be fastened to prevent chafing on abrasion. All wiring shall be color or number coded, and the coding scheme shall be included on the electrical diagrams. All wires shall be adequately sized, and all circuits except the ignition shall have overload protection. All electrical connections fastened with terminal nuts shall be equipped with lock washers or nut locking devices. The electrical system circuitry, including all components, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-183, Composition I or II, as applicable with a one percent copper 9-quinolinolate (by weight) based on the nonvolatile content of the varnish. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated. Components of circuit elements whose functions will be adversely affected by the varnish coating shall not be treated. When used, the varnish shall be applied by spray, brush or a combination of both to give a minimum dry film thickness of one mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.19.1 Lights. Each truck shall be equipped with a minimum of four sealed beam CIM floodlights confirming to SAE J598, PAR 46, Trade No. 4478, which shall be shock mounted in elastomer ring housings. Each light shall be adjustable a minimum of 45 degrees above and below horizontal plane. Two lights shall be adjustable to illuminate a minimum of 180 degrees to the front of the truck, and two lights shall be adjustable to illuminate 180 degrees to the rear of the truck. Adequate protection against damage of each light, by means of position or guard, shall be provided. The two front floodlights shall be mounted on the outside of both uprights.

3.19.2 Taillight. One automotive type combination stoplight and taillight shall be furnished. The taillight shall conform to SAE designations as specified in SAE J759. The lamps shall be either sealed units mounted in elastomer rings or bulbs and lens mounted in elastomer housings. If furnished, the bulb shall be Trade No. 1157. The taillight shall conform to SAE-STI, red minimum 3-inch diameter, and shall be recessed or

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mounted on the rear and within the plan outline of the truck, and shall be protected against damage by a guard.

3.19.3 Horn. The truck shall be equipped with an electric horn with a push-button mounted in the center of the steering wheel. The horn button assembly and electrical wiring for the horn shall be constructed to be moisture and weather resistant to prevent entry of moisture when operated or stored outdoors under all weather conditions.

3.20 Operator's overhead guard. An overhead guard with falling objects protective structure (FOPS) shall be furnished and shall be in accordance with ANSI B56.1.

3.21 Personnel cab. When specified (see 6.2), the truck shall be equipped with an insulated, weathertight personnel cab. The cab clearances shall be in accordance with SAE J154. The cab shall inherently resist torsional strain and withstand all operational shock and vibration under the operating conditions specified herein. The cab shall be constructed of metal. The cab shall be equipped with a minimum of two fully opening or removable doors. Each door shall have a safety latch, opening stop and hold-open catch. The cab door hold-open catches shall be mounted to rigid points on the truck. All hardware shall be of the heavy-duty truck or industrial type, resistant to corrosion. The cab shall be equipped with not less than two ventilation fans not less than 6 inches in diameter. Fans shall be directionally adjustable to direct airflow toward all glazing. Fan(s) shall have "on-off" switch(es). There shall be dome light in the ceiling of the cab controlled by a separate switch mounted on the instrument panel. The cab shall be removable and independent from the overhead guard (3.20). When the cab is removed the guard shall meet the requirements of ANSI B56.1, including FOPS. The overall height to the top of the cab shall be not more than 95 inches.

3.21.1 Glazing. The cab shall provide the operator with all around visibility. The total glazed area shall be no less than 4000 square inches of tinted windows consisting of windshield, sliding or roll-down door windows, rear window, roof window, and other windows as determined by cab design. All glazing shall conform to ANSI Z26.1, Type AS1 in the windshield, rear window, and roof window and Type AS1 or AS2 in other windows. Fixed glazing shall be sealed in waterproof stress-absorbent molding, and movable windows shall be sealed on extra heavy type, pile lined channels.

3.21.2 Interior. The interior of the cab shall be fitted with insulation, retarding both heat loss and noise intrusion. The insulation and framing shall be covered with an interior skin and headliner, Providing a smooth, cleanable protective cover.

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3.21.3 Windshield and window wipers. The cab shall be equipped with at least three powered-wiper assemblies to clear the windshield, the rear window and the roof window. The wiper motors shall provide not less than 90 inch-pounds of torque when electric motors are furnished, or provided not less than 170 inch-pounds of torque when air motors are furnished. Wipers less than 179 inch-pounds of torque when air motors are furnished. Wipers shall be capable of variable stroke sped up to at least 60 strokes per minutes. Self-parking features shall be provided. Wiper arms and blades shall be of the commercial heavy-duty type having saddle and pin blade fastenings. The wiper arms shall provide tension on the glazed surface of at least 1 ounce for each inch of wiper arm length. Air motors shall be plumbed to exhaust air outside of the cab. The operator controls shall be installed on or adjacent to the instrument panel.

3.21.4 Heater. A recirculating-air type heater shall be furnished. The heater shall have a rating of at least 18.000 Btu/hours. The heater controls shall be near the operator and shall provide for variable selection of blower speeds. Unless Type A winterization (see 3.26.1) is specified, the heater shall be capable of maintaining 40 degrees F at floor level in ambient temperatures of -20 degrees F. The heater, either by itself or by ducts to the windows, shall be capable of removing all ice from all windows within 30 minutes after engine start at an ambient temperature of -25 degrees F.

3.21.5 Exhaust system. All nonturbocharged diesel engines shall have a spark arrestor, either separate or incorporated into the muffler. In either case, the arrestor shall be capable of being cleaned without removing any components, except the exhaust shield, if furnished. Exhaust gases shall be emitted vertically above the operator's head or at the rear or side rear of the truck within the plane outline of the truck. The exhaust system shall prevent entry of rain and shall be shielded as necessary to prevent burns to the operator and maintenance personnel while performing scheduled maintenance or standing on the step surfaces of the truck.

3.22 Performance.

3.22.1 Lifting speed. Speed of the lift with rated load shall be not less than 55 feet per minute over the entire distance from ground level to maximum list height.

3.22.2 Lowering speed. Speed of lowering forks with rated load shall be not more than 80 feet per minutes over the entire distance from maximum lift height to ground level. The hydraulic control system shall be damped, metered or manually regulated to reduce shock and prevent the truck from overturning when rated load is lowered at maximum speed and then stopped quickly.

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When nuclear safety is specified (see 6.2), maximum lowering speed, immediately before forks contact ground, shall be not more than 65 feet per minute. The unloaded forks shall lower the entire distance from maximum lift height to ground level at a speed of at least 35 feet per minute.

3.22.3 Right angle turn. Trucks, when carrying rated capacity load, shall be capable of backing through a turn in either direction and returning to the same or a parallel position within dimension "A" of figure A-1 when demonstrated as specified herein.

3.22.4 Speed. While carrying the rated load on a level surface, the truck shall be capable of attaining a minimum of 15 miles per hour in both the forward and reverse directions.

3.22.5 Slope ascension, first speed, forward direction. Trucks shall be capable of ascending a 20 percent grade on a dry concrete surface with rated capacity load. Trucks carrying the rated load shall be able to accelerate from a dead stop on this slope. The truck shall drive, from a horizontal surface, up a 20 percent ramp that is longer than the forklift and onto another horizontal surface.

3.22.6 Upright tilt. When unloaded, truck masts shall have at least 3 degrees forward tilt, and at least 10 degrees rearward tilt. Positive means shall be furnished to prevent cavitation of the tilt cylinders at minimum 120 degrees F oil temperature when tilting rated load to full forward tilt at engine low idle speed. When nuclear safety is specified (see 6.2), the hydraulic tilt circuit shall be provided with a locking valve arrangement to prevent uncontrolled tilting in the event of a hydraulic line failure.

3.22.7 Upright height. Collapsed mast height of the unloaded truck measured from the ground to the top of the uprights in a vertical position shall not exceed that specified in the applicable specification sheet.

3.22.8 Maximum fork height. The maximum fork height of a truck bearing rated capacity load and the uprights fully extended, measured from the ground to the load carrying surface of the fork at the rated load center, shall be not less than that specified in the applicable specification sheet.

3.22.9 Free lift height. Height of forks with rated load, without increase in the specified collapsed mast height shall be not less than that specified in the applicable specification sheet.

3.22.10 Underclearance. Minimum underclearance beneath the mast assembly, when in true vertical position, shall be not less

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than that specified on the applicable specification sheet when the truck is fully loaded. When the truck is unloaded, the underclearance of the truck other than the mast and axles, shall be not less than 7 inches. Hydraulic fittings, hoses, tubing, and linkages shall not be the lowest portion of the truck and shall be protected by structural members from striking obstacles.

3.22.11 Drift. The lift assembly shall be capable of holding the rated load at maximum height not less than 10 minutes with not more than 1.0 inch of vertical drift and not more than 1 degree of rotational drift from the vertical.

3.22.12 Stability. Trucks shall meet "forward stacking", "forward travel", "lateral stacking", and "lateral travel" stability requirements of ANSI B56.1.

3.22.13 Truck operations. The truck shall complete 6 to 8 circuits per hour, for a cumulative total of 40 hours, when demonstrated as specified in 4.4.11. The operational periods shall be at least nine hours long with 1 1/2 hour for operator's breaks and lunch. During this nine hour period, no maintenance is allowed. Only routine maintenance as prescribed by the manufacturer is allowed and only at the end of the nine hour operational period. The truck shall show no signs of unusual or detrimental wear. No failure shall occur during the 40 hour operational test. A failure is defined as any malfunction that can not be corrected within 30 minutes by adjustment, repair, or replacement, which may cause cessation of operation, degradation of performance capabilities below specified levels, damage to the truck by continued operation, or personnel safety hazard. At the completion of this test, the truck shall be capable of meeting all the performance criteria setforth herein, however, testing previously accomplished during the First Article evaluation does not have to be repeated.

3.22.14 Noise.

3.22.14.1 Noise limits. The noise level of the truck (excluding horns and alarms) shall not exceed 85 dB(A) when tested in accordance with 4.4.16.1 and 4.4.16.2.

3.22.14.2 Noise hazard. When the noise level of the truck exceeds 85 dB(A) when tested in accordance with 4.4.16.1 and 4.4.16.2, noise hazard caution signs conforming to MIL-STD-1474 shall be posted by the contractor. The contractor shall also insert into the maintenance manual a discussion of the noise hazard, the requirement for hearing protection, the noise level of the truck at the operator's station, and the distance (contour line) from the truck at which 85 dB(A) will always be met when tested in accordance with 4.4.16.3.

3.23 Preparation, primer, and paint.

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3.23.1 Preparation and primer. All external surfaces and all surfaces normally painted shall be prepared and primed using a procedure compatible with the final paint.

3.23.2 Forest green. Unless otherwise specified (see 6.2), the truck shall be primed with MIL-P-23377, MIL-P-52192, MIL-P-53022Ø or MIL-P-53030 and painted with polyurethane paint conforming to MIL-C-83286, Color Number 24052 of FED-STD-595. All exterior trim (wheels, radiator grill, rims on lights, etc) shall be the same type and color as the exterior paint.

3.23.3 Desert sand. When specified (see 6.2), the truck shall be painted with polyurethane paint specified in 3.23.2, except it shall be Color Number 30313 of FED-STD-595. All exterior trim (wheels, radiator grill, rims on lights, etc.) shall be the same type and color as the exterior paint.

3.23.4 Lusterless gray. When specified (see 6.2), the truck shall be painted with polyurethane paint specified in 3.23.2, except it shall be Color Number 36081 or FEB-STD-595. All exterior trim (wheels, radiator grill, rim on lights, etc.) shall be same type and color as the exterior paint.

3.23.5 Walkway coating. Floor plates and step surfaces shall be coated with a nonslip coating compound or be furnished with a nonslip metal surface.

3.24 Identification marking. All identification, instruction and marking plates shall conform to MIL-P-514, type I, style I, composition C material. All plates shall be securely attached by screws, bolts or rivets in a conspicuous location, and shall be furnished and mounted by the contractor. The truck shall be identified with an identification plate located in a visible and safe location on the instrument panel or firewall of the truck. The identifying data to be applied to the identification plates shall be as indicated herein as defined in MIL-STD-130. The truck shall also be equipped with instruction, warning, and caution plates conforming to MIL-P-514, type III, and describing any special or important procedure to be followed in assembling, operating, loading, and servicing of the truck and its components.

3.24.1 Shipping data plate. A shipping data plate conforming to MIL-P-514, type III, shall be furnished and in addition shall show the silhouette of the truck in transport position indicating the center of gravity. Wheel loading information specified in 3.24.2 may be combined in one plate with the requirements specified herein.

3.24.2 Wheel loading plate. Each truck shall be equipped with a wheel loading plate conforming to MIL-P-514, type III. As

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a minimum, the plate shall have the following information:

Wheel loading (no load on forks)
Drive wheels (each) pounds
Steer wheels (each) pounds

Wheel loading (gross vehicle weight)
Drive wheels (each) pounds
Steer wheels (each) pounds

3.25 Truck marking. Markings for vehicle painted forest green shall be black paint, Color Number 37038 of FED-STD-595. Markings for trucks painted desert sand shall be white paint, Color Number 37875 of FED-STD-595. Forklifts painted lusterless gray shall have no markings. Location information, and character size shall be as follows:

Information	Example	Location	(Height)
Agency	US	On each side	1-1/2 inches
Registration Number	87X315	On each side	1-1/2 inches
Capacity	10,000 - CAP	On each side of mast	2 inches
Safety	No riders	On rear of mast	2 inches
Tire pressure	TPXX	On frame above each tire	1 inch
Fuel fill rate	Max. safe fuel accept - gpm	Near fuel tank filler	1 inch

3.26 Winterization. When specified (see 6.2), the truck shall be winterized as follows:

3.26.1 Type A. Type A winterization shall protect to -65 degrees F, and shall consist of the following components:

a. Power plant heater. Coolant, engine oil and battery heaters shall be provided. Heaters shall operate on 110-volt, alternating current, and shall be wired through a junction block to a single three pronged (male), weather proof, slave receptacle for receiving external power and grounding vehicle. A three wire connecting cable, 25 feet long and of adequate line capacity to supply power for heater units simultaneously, shall be furnished. Connecting cable shall include a matching female connector at the vehicle and a standard, weatherproof, three pronged (two power

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plus one ground) male connector at the other end. Electrical apparatus shall conform to Motor Carrier Safety Regulation 393.77(c)(7). Electrical insulation, of connecting cable, shall withstand normal operating stresses in low ambient air temperature (down to minus 65 degrees F) without loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. A carrier for the connection cable shall be mounted within the cab or engine compartment and shall provide positive cable retention during vehicle operation. Heaters shall be furnished as follows:

(1) Coolant heater, 1500-watt minimum rating, shall be installed in the engine block or lower coolant inlet hose. A coolant circulating pump, driven by a 110-volt alternating current motor, shall be provided.

(2) Immersion type engine oil heater, 300 watt minimum rating with 190 to 195 degrees F thermostat, shall be installed in oil pan.

b. Battery. Heavy duty (lead-acid) battery shall be furnished. The battery shall be heated by an electrical heater. Battery heater shall have a capacity adequate to maintain the battery electrolyte at a temperature of not less than 10 degrees F during vehicle exposure in ambient temperature as low as minus 65 degrees F, and shall embody a thermostat to limit the temperature of the electrolyte to not more than +80 degrees F.

c. Hard top cab. Cab shall be bolt on unit type and removable as specified in 3.21. a floor covering is required on thermal and acoustical insulation. Boots are required for insulation to close all openings where attachments and controls enter the cab.

d. Thermostat. Install a 170 to 190 degrees F engine thermostat.

e. Personnel cab heater with defroster. A universal hot water heater and defroster shall be installed. The heater shall be of sufficient capacity to maintain a temperature of +40 degrees F at cab floor level in an ambient temperature of -50 degrees F. Use may be made of sufficient thermal insulation material in the cab area to insure that the cab heater can meet these temperature conditions.

f. Air drier kit. Air dryer kit shall be included if air brakes are furnished.

g. Arctic type lubricants. Engines, gearboxes, transmissions and other reservoirs shall be compatible with arctic type lubricants.

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h. Antifreeze. Antifreeze shall comply with either Military Specifications MIL-A-11755 or MIL-A-46153 or shall be standard commercial arctic type. Coolant liquid shall be protected to at least -65 degrees F.

3.26.2 Type B. Type B Winterization shall protect to -25 degrees F and shall consist of the following components.

a. Hard top cab. A cab conforming to 3.26.1.c shall be included.

b. Antifreeze. Antifreeze shall comply with either Military Specifications MIL-A-11755 or MIL-A-46153 or shall be standard commercial type. Coolant liquid shall be protected to -25 degrees F or lower temperature.

c. Thermostat. Install a 170 to 190 degrees F engine thermostat.

d. Air drier. Air drier shall be included if air brakes are furnished.

e. Personnel cab heater with defroster. A universal hot water heater and defroster shall be installed. The heater shall have sufficient capacity to maintain a temperature of +40 degrees F at cab floor level in an ambient temperature of -25 degrees.

3.27 Air transportability. The forklift shall be air transportable in C-130, C-141, and C-5 Aircraft without removal of any components (including cab, if so equipped) by driving on and off the aircraft, from and to ground level. The truck shall meet or exceed all requirements of AFSC Design Handbook DH 1-11. In addition, the fully operational forklift, with no components removed and with all fluid operating levels, including the hydraulic system at maximum and the fuel tank at least 1/4 full, shall have a maximum per axle load of 13,000 pounds.

3.28 Slinging and tiedown provisions. Permanently affixed slinging and tiedown provisions that enable the truck to be lifted in its normal position and to be fastened to the floor or deck of a transportation medium shall be provided. Slinging and tiedown provisions shall be in accordance with MIL-STD-209. Tiedowns shall be located and be of sufficient strength to meet the gravitional ("G") loads specified in section 2B2 of AFSC Design Handbook DH 1-11. For location of tiedown points in C-130, C-141 and C-5 aircraft see DH 1-11. If practical, the slinging and tiedown points shall be the same. The notation "LIFT HERE" shall be stenciled in the color marking as specified in paragraph 3.24 near each lifting point. Center-of-gravity location (both horizontal and vertical) in the air transport mode and tiedown instructions in the C-130 and C-141 Aircraft shall be on the forklift in accordance with MIL-A-B421. Each tiedown

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point shall be marked with its capacity; for example: "TIEDOWN, 10,000 POUNDS."

3.29 Structural and hydraulic. The entire forklift assembly shall be capable of withstanding three times the rated load in static condition (see 4.4.20).

3.30 Electromagnetic interference (EMI). The truck shall comply with the EMI requirements of MIL-STD-461B, Class C1, Group II, Figure 8-1, except the emission limits for UM03 (Figure 8-1) shall be relaxed by 20dB for all frequencies.

3.31 Workmanship. The workmanship shall be high quality when compared to industry standards. There shall be no rough surfaces or edges on machined or flame cut pieces. All adjacent permanently installed and hinged metal panels shall be aligned and evenly spaced with gaps not more than 1/16 of an inch.

3.32 Welding. All welds shall be smooth, continuous, with complete heat penetration and shall transmit stress without permanent deformation or failure when the truck is subjected to normal service loading or the testing in Section 4 of this specification. The contractor's welders, welding operators and welding procedures shall be qualified to meet the requirements of MIL-STD-248.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herin. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herin, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it

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commit the Government to acceptance of defective material.

4.2 Classification of Inspection. Inspections shall be classified as follows:

a. First article truck evaluation and demonstrations (see 4.3 and 4.4).

b. Production truck inspections (see 4.5).

c. Inspection of packaging (see 4.7).

d. User test (see 4.8).

e. In-production (random sample) test (see 4.9).

4.3 First article evaluation and demonstrations. Prior to delivery, the manufacturer shall make one truck available for first article evaluation and demonstrations. The truck and its components shall be compared with this document, published literature, calculations and test data to verify compliance with this specification. The truck shall then be subjected to the demonstrations specified herein. Occurrence of any of the following conditions during the testing specified in Section 4 of this specification shall constitute failure of the first article evaluation and shall be cause for rejection of the truck:

a. Inability to meet specified performance requirements or inability to complete any test.

b. Permanent deformation, overheating, malfunction or leakage of fuel, hydraulic oil or lubricants.

c. Any binding, jerking or uneven operation of any component or system.

d. Any wearing, galling, gouging or other removal of material that, in the opinion of the Government Contracting Officer, will cause premature failure.

4.3.1 Preparation for demonstration. Prior to demonstration (see 4.4), the following shall be performed:

a. Service the truck with lubricants and fuel that is compatible with the ambient temperature. The hydraulic fluid shall be the type specified herein.

b. Set the hydraulic relief valve as specified by the manufacturer and record. Any change made to this setting,

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except for the fork lifting overload test (see 4.4.28) hydraulic pump test (4.4.34), and hydraulic cylinder test (4.4.37), shall be cause for failure of all tests.

c. Run in truck as required.

4.4 DEMONSTRATIONS

4.4.1 Demonstration conditions. Unless otherwise specified herein, demonstrations shall be conducted at the ambient temperature and climatic conditions existing at the place of demonstration. Only that maintenance established by the contractor and submitted as a maintenance schedule prior to commencement of the demonstration shall be performed during the demonstration.

4.4.2 Rated load. The rated load shall be a 10,000 (plus 200, minus zero) pound load, 96 (plus 1/2, minus zero) inches on each side. The cube shall be provided by the contractor.

4.4.3 Maximum lift height, collapsed mast height, and free lift height. Demonstrate the truck in accordance with demonstration method No. 1. Nonconformance to 3.22.7, 3.22.8, or 3.22.9 shall constitute failure of this demonstration.

4.4.4 Lifting speed and lowering speed. Demonstrate each truck in accordance with method No. 2. Nonconformance to 3.22.1 or 3.22.2 shall constitute failure of this demonstration.

4.4.5 Rightangle turn. Demonstrates the truck in accordance with demonstration method No. 3. Dimension "A" greater than specified in the applicable specification sheet shall constitute failure of this demonstration (see 3.22.3).

4.4.6 Travel speed. Demonstrate each truck in accordance with demonstration method No. 4. Nonconformance to 3.22.4 shall constitute failure of this demonstration.

4.4.7 Slope ascension and parking brake. Drive the truck forward, with load, onto a 20 percent ramp that is at least 10 feet longer than the truck. Stop the forklift on the ramp and using only the parking brake, hold for a least 60 seconds. Release the brake and drive up the ramp onto a second horizontal ramp. There shall be no contact (except by tires) between the forklift and the ramp or either horizontal surface. Starting on the lower horizontal surface, drive the forklift in reverse, with rated load, to about mid point up the ramp. Hold the truck for 60 seconds using only the parking brake.

4.4.8 Tilt. Demonstrate the truck in accordance with demonstration method No. 6. Nonconformance to 3.22.6 shall constitute failure of this demonstration.

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4.4.9 Drift. Demonstrate the truck in accordance with demonstration method No. 7. Nonconformance to 3.22.11 shall constitute failure of this demonstration.

4.4.10 Stability. Conduct "forward stacking", "forward travel", "lateral stacking" and "lateral travel" tilting platform demonstrations in accordance with ANSI B56.1. Failure to meet the requirements specified in ANSI B56.1 shall constitute failure of this demonstration.

4.4.11 Truck operations. Demonstrate the truck in accordance with demonstration method No. 8. Nonconformance to 3.22.13 shall constitute failure of this demonstration.

4.4.12 Parking brakes. The parking brake, or when an emergency braking system is provided for nuclear safety as specified in 3.11.2, stopping distance shall be demonstrated on a level surface in accordance with ANSI B56.1. Nonconformance to 3.11.2 and ANSI B56.1 shall constitute failure of this demonstration.

4.4.13 Starter Disconnect. Start the truck by energizing the starter switch. Release starter switch and immediately reactivate starter switch while engine is operating. Any evidence of starter engagement while engine is running shall constitute failure of this demonstration.

4.4.14 Load securing devices and carriage assembly. When load securing devices are provided, the devices and carriage assembly shall be tested for verification of specified requirements. Nonconformance to 3.9 shall constitute failure of this demonstration.

4.4.15 Side shift. Operate the side shift attachment for 100 full cycles at full speed from extreme left to extreme right and back to extreme left with the rated load. During the last 50 cycles, the control shall be released abruptly when the load is in approximately midposition between the left and right extremes and then the side shifter shall be restarted. Nonconformance to 3.9.3 shall constitute failure of this test.

4.4.16 Noise level measurement.

4.4.16.1 Operator's station. Measure the maximum steady state noise levels of the truck in accordance with the procedures of SAE J919, except (see 3.3):

Lifting rated load at maximum lift speed.
Engine stalled against converter.

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Engine stalled against hydraulic system relief.
Governed engine speed, no load.
Maximum acceleration while carrying rated load.

Nonconformance to 3.22.14.1 shall constitute failure of the test. Any level in excess of 85 dB(A) shall invoke the requirements of 3.22.14.2.

4.4.16.2 Exterior. Measure the maximum steady state noise level in accordance with SAE J88, except the distance shall be 7.5M in lieu of 15M. Nonconformance to 3.22.14.1 shall invoke the requirements of 3.22.14.2.

4.4.16.4 Contour line. When required (3.22.14.2), the 85 dB(A) contour line shall be determined for the stationary mode of operation specified in 4.4.16.1 at which the highest noise level was recorded. Measurements shall be taken normal to the four major surfaces (front, rear, and sides), at their centerline, and under the conditions specified in 4.4.16.1 except the microphone shall be moved to the distance at which an 85 dB(A) level is obtained. The distance shall be recorded and used to generate the contour line.

4.4.17 Transmission neutral switch. With the engine not running, place transmission selector in lowest forward gear and energize starter switch. Repeat for all forward and reverse gears. Any evidence of starter engagement while transmission is in gear shall be cause for rejection.

4.4.18 Fork visibility. With no load on the forks, test for compliance with the visibility requirements of 3.9.

4.4.19 Axle weight. Weigh front and rear axle with the forklift fully operational, with the cab installed, and at least 1/4 tank of fuel. No axle shall weigh more than 13,000 pounds.

4.4.20 Overload. Place truck on level surface and support it with blocks under the axle or frame to relieve the load on the tires. Secure the truck to the floor in a manner to compensate for the additional test load. Gradually apply load to the forks until three times the rate capacity (30,000 pounds at 48 inch horizontal and vertical load center) is on forks. Leave this load on the forks for at least 10 minutes. Remove load and inspect truck structure, frame, mast assembly and hydraulic system for deformation, cracks, broken welds, hydraulic system leaks, etc. Inspect forks at least six inches each side of heel by magnetic particle or dye penetrant method.

4.4.21 Fail-safe hydraulic lift system. Raise the rated load four to seven feet above ground level with the mast approximately vertical. Measure vertical distance to one fork tip. Disconnect a hydraulic line that supplies pressure to the

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lift system to simulate a line failure. When fluid stops draining, remeasure vertically to same point on fork tip. A drop of more than 1/2 inch shall be cause for rejection. Measure the mast tilt. Disconnect the hydraulic lines that supply pressure in both directions, to all tilt cylinders. When fluid stops draining, measure mast tilt. A change of more than one degree shall be cause for rejection. Lower the forks to ground level using the override system.

4.4.22 Service brakes. Perform the stop test of ANSI B56.1, except the maximum brake pedal force during the test shall be 80 pounds.

4.4.23 Steering. With no load on the forks, engine at idle speed and truck stationary, measure the tangential force on the steering wheel necessary to cause the truck steering wheels to turn. Count the number of turns required to go from full right to full left. Failure to meet the requirements of 3.11 shall be cause for rejection.

4.4.24 Overhead guard. Test the overhead guard in accordance with ANSI B56.1. Nonconformance to 3.20 shall be cause for rejection.

4.4.25 Slinging and tiedown provisions. Test the slinging provisions by applying the load for the time period specified in MIL-STD 209. Test the tiedown provisions by applying the ultimate load for the time period specified in MIL-STD-209. Nonconformance to 3.28 shall constitute failure of this test.

4.4.26 Load securing devices. Apply a total forward load of 30,000 pounds divided equally between all load securing devices (see 3.9). The entire 30,000 pounds does not have to be applied at one time, the maximum load on each device may be applied independently. Any deformation, separation, broken welds, etc, shall be cause for failure.

4.4.27 Fork lifting mechanism. The fork lifting mechanism shall be operated through 100 complete continuous cycles (from ground level to maximum lift height and back to ground level) with no load and 200 cycles with rated load on the forks. Any failures of lift mechanism components, overheating of engine coolant system or hydraulic system, inability to complete the test of any other failure shall be cause for rejection.

4.4.28 Fork lifting overload. With 125 percent of the rated load (12,500 pounds at 48 inch horizontal and vertical load center) on the forks and with the truck tied down to prevent overturning, the lifting mechanism shall be operated through 25 complete continuous cycles. From ground level to maximum lift height and back to ground level shall be considered one cycle. Perform at least 25 continuous cycles. Side shift the 12,500

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pound load from extreme right to extreme left and back to extreme left. This shall be considered one cycle. Perform at least 25 continuous cycles. Occurrences of any conditions listed in 4.3 shall be cause for rejection.

4.4.29 Fork spacing mechanism. The fork spacing shall be operated with no load on the forks through 1ØØ complete continuous cycles. A cycle shall consist of starting at closed position, going to open position and returning to closed position. Nonconcormance with 3.9.2 or any evidence of binding, galling, erratic movement, etc, shall be cause for rejection.

4.4.3Ø Electromagnetic interference (EMI). Test the truck to verify it meets the EMI requirements of 3.3Ø. Failure to meet these requirements shall be cause for rejection.

4.4.31 Maintenance operations. Perform the maintenance operations described in 3.5.1. Failure to complete any task in the specified time shall be cause for rejection.

4.4.32 Cold starting. Place the truck in an ambient temperature of -25 degrees F, or cooler, until all components are established at the ambient temperature. A fine mist of water shall be applied to all windows to form a film of ice. The engine shall start within five minutes and rum smoothly at idle speed without continued engine control adjustment within 15 minutes. At this time all travel, braking, and load handling mechanisms, shall be operated. The forks shall be raised to maximum lift height, and sideshifted and adjusted laterally between extreme positions. Starting aides such as the ether injection or glow plug furnished with the truck may be used to assist starting: however, no external power or heat source shall be used. The heater/defroster shall completely clear all windows of ice and water vapor within 3Ø minutes after engine start. The test chamber shall maintain the ambient temperature at -25 degree F, or colder, during the entire test. Failure to meet any of these requirements shall be cause for rejection.

4.4.33 Hydraulic system vacuum. Install vacuum gauge into the pump suction line immediately adjacent to the pump or at the pump. Install a thermocouple in the hydraulic reservoir. Run the pump at governed RPM. Measure the pump inlet pressure in inches of mercury vacuum at the specified temperature and speed. Nonconformance to 3.8.5 shall constitute failure of this test.

4.4.34 Hydraulic pump(s). Operate each pump at maximum governed speed and at 1 1/2 times maximum system working pressure for not less than two minutes. Check for external leaks at assembled surfaces and the shaft seal. Nonconformance to 3.8.6 shall constitute failure of this test.

4.4.35 Hydraulic filter. Block filter housing outlet port.

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Pressurize the housing to at least 100 psi (30 psi for suction filters) and hold for not less than 60 seconds. Examine for leaks. Install a new filter element. Establish maximum system flow through the filter at a fluid temperature of 125 to 155 degrees F. Measure and record the pressure drop across the filter assembly. Plug the filter element chamber fluid exit, or otherwise obstruct all flow through the filter element. With the filter housing outlet post open and the fluid at 125 to 155 degrees F, set inlet pressure to 10 psi (1.75 psi for suction filters). Increase inlet pressure until the contamination indicator enters bypass. Record this pressure.

4.4.36 Hydraulic filter life. Hydraulic filter life test shall be conducted on the truck(s) during the reliability test (see 4.4.11). At the conclusion of the reliability test, establish maximum system oil flow through the filter at an oil temperature of 150, plus or minus 5 degrees F, and note if the element is bypassing. Nonconcorfmance to 3.8.7 as evidenced by indicator reading in bypass shall constitute failure of this test.

4.4.37 Hydraulic cylinders. Test each hydraulic cylinder in accordance with Test Method Number 9. Nonconformance to 3.8.8 shall constitute failure of this test.

4.4.38 Hydraulic system cleanliness. Test truck in accordance with Test Method Number 10. Nonconformance to 3.8.9 shall constitute failure of this test.

4.4.39 Hydraulic Fittings. The contractor shall furnish a complete set of hose assemblies with reusable fittings for this test. Remove one hose assembly that has permanent type fittings. Replace the hose assembly with an equivalent hose assembly with field attachable fittings. Repeat until each hose assembly with permanent type fitting is replaced with hose assembly with reusable fittings. Nonconformance to 3.8.10 shall constitute failure of this test. Reinstall the original hose assemblies on the truck.

4.4.40 Post test disassembly and examination. Upon completion of all tests measure and record the compression in each engine cylinder and dismantle the truck sufficiently to permit detailed disassembly and visual inspection of the following components:

a. Engine cylinder heads, all halves, two pistons and piston pins taken from the two cylinders having the lowest compression readings at the end of the test, one main bearing and one connecting rod bearing.

b. Torque convertor and transmission.

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c. Universal joints and drive shafts.

d. Differential.

e. Front axle bearings and seals.

f. Complete brake system components.

g. Steering system cylinders, valves, and pump.

h. Hydraulic system pump, valves, one cylinder of each size, control levers, and any hose and fitting where leakage occurred during the tests.

i. Steer axle bearings and seals.

j. Mast assembly bearings and chains.

Additional disassembly and examination shall be made at the option of the contracting officer. Record complete description of the condition of all parts and assemblies examined and take pictures of any dificiency. Each evidence of corrosion which affects the functions of the truck, each evidence or permanent deformation or breakage, or excessive wear of any component shall constitute a defect. Manufacturing tolerances listed in the contractor's quality control program shall be made available upon request and shall be used as the basis for determining excessive wear.

4.5 Production run-in. Each truck carrying a rated load, shall be subjected to a production run-in and servicing as specified herein. Inability to complete the run-in, overheating failure of any components, malfunction of any control, or evidence of leakage of fluids shall be cause for rejection of the truck by the Government until such deficiencies are corrected.

4.5.1 Travel. Demonstrates starting, ascending and descending a 20 percent ramp, steering, inching and high speed travel. Operate all lights, horn, and brake system not less than two times during this demonstration.

4.5.2 Mast assembly. Demonstrate ability to lift rate load to full lift height at full lift speed and lower load at full opened valve without exceeding maximum lowering speed. Demonstrate ability to tilt mast full forward and full rearward with rated load on the forks. Demonstrate ability to side shift rated load throughout full range of side shift. Demonstrate ability to feather hydraulic controls for control of all hydraulic functions. Each operation shall be performed at least three times.

4.5.3 Servicing and adjustment. Unless otherwise specified (see 6.2), prior to delivery, the manufacturer shall service and

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adjust each truck for customer operational use, including at least the following:

(a) Adjustment of mast assembly.
(b) Adjustment of engine, transmission, and power train.
(c) Adjustment of brake system.
(d) Adjustment of steering.
(e) Continuity check of the electrical system, including filling and charging of battery.
(f) Check torque on all wheel lug nuts and proper inflation of tires.
(g) Service the cooling system.
(h) Check fluid levels and fill to required level if necessary.

4.6 First article demonstration report. The contractor shall prepare and submit the first article demonstration report to the contracting officer for approval within the time frame specified (see 6.2). The report shall include a detailed description of the examination and demonstration performed, deficiencies encountered and actions taken to correct the deficiencies.

4.7 Inspection of packaging. Each truck, when completely prepared for shipment, shall be inspected for the preservation, packaging, and marking as specified in the contract. Nonconformance shall be cause for rejection of the packaging.

4.8 User test. The user test sample shall be delivered to Little Rock AFB AR for 45 available days (seven days a week) of user tests. The 45 available days shall be consecutive, except when interrupted for maintenance/repair downtime, as defined herein. The user test will not begin until after compleiton of the first article test, except the EMI and environmental test, of the contractor test sample as defined by 4.4.3Ø and 4.4.32. Production run-in (see 4.5) and the examination of product (see 4.3) shall be performed on the user test sample prior to shipment of the user test sample to Little Rock AFB. The user test sample shall not be shipped to Little Rock AFB without prior 15 day notification to the PCO. The test period shall end when 45 days of availability are complete or 6Ø calendar days have elapsed, whichever occurs first. If the user test sample has not been available for 45 test days after 6Ø calendar days have elapsed, the user test shall be considered a failure. The vehicle shall be available for any one period of eight consecutive hours during each day of the 45 day test. The periods shall be scheduled by the Little Rock AFB representative and the schedules shall be provided to the contractor at least 12 hours in advance. Any period of more then four hours of the same day that the vehicle is out of service for maintenance or repair shall not be considered an available day for the test period.

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The vehicle shall be operated, and scheduled maintenance procedures contained in the preliminary technical data to be supplied with the test sample. The preliminary technical data shall contain procedures for preparation/tiedown of the forklift for air shipment. The contractor will provide any special tools/ equipment required to perform operators maintenance/scheduled maintenance. Any parts or corrective actions required shall be provided by the contractor. Any maintenance actions performed by the contractor shall not begin before notification to a Little Rock representative, and Government personnel shall be allowed to observe their actions. The contractor shall provide all forklift consumables required in the course of the test except for diesel fuel, water, compressed air and electrical power which will be furnished by Little Rock AFB. The contractor shall provide on-site familiarization to the operators and mechanics as necessary to perform this test. Test operations will be used in or simulate normal Air Force usage and will not exceed any performance requirements defined by Section 3 of this specification. Nonconformance with the performance requirements of this specification shall be cause for rejection. The contractor shall maintain a daily log. The log shall be signed by the Little Rock AFB representative and the contractor after each day of activity. The daily log shall document, but not be limited to, the following:

a. Operating time.

b. Tasks completed.

c. Technical manual errors.

d. All failures and corrective actions taken.

e. Operational problems encountered.

f. Little Rock AFB representative comments.

The contractor shall report the user test results, including daily logs of activities, in the first article inspection report.

4.9 In-production (random sample) test. Any vehicle from the production line may be selected by the procuring activity at any time during the production phase of the contract to be subjected to the test described in 4.9.1. Any deficiencies found as a result of this test shall be considered prima facie evidence that all trucks already accepted are similarly deficient. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on units previously produced and accepted under the contract, as well as on the forklift used for this test and on all vehicles to be produced during the remainder of the contract.

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4.9.1 In-production test. The contractor is responsible for the performance of the examination in 4.3 and all testing in 4.4 at his own or other facilities approved by the Government.

a. Maximum lift height and collapsed mast height (4.4.3).

b. Lifting speed and lowering speed (4.4.4).

c. Right angle turn (4.4.5).

d. Travel speed (4.4.6).

e. Slope ascension and parking brake (4.4.7).

f. Tilt (4.4.8).

g. Stability (4.4.10).

h. Truck operations (4.4.11).

i. Axle weight (4.4.19).

j. Service brakes (4.4.22).

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified (see 6.2), each complete truck and all items issued with the truck shall be preserved and packed for retail distribution directly to a using customer and in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with carrier rules and regulations applicable to the mode of transportation. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coating, volatile corrosion inhibitor, or desiccated packs. Items requiring protection from physical damage to surfaces or which are fragile by nature shall be protected by wrapping, pack compartmentalization, or cartonizing of the individual item. The cooling system shall be filled with a fresh clean solution of 50 percent water and 50 percent ehtylene glycol type antifreeze with rust inhibitor, and a tag attached indicating the temperature to which the cooling system can be subjected to before freezing. The crankcase shall contain the required amount of lubricating oil. A guide shall be furnished indicating the preservation and packing materials to be removed, in detain, and any other servicing required prior to placing the truck in operation. Marking shall reflect as a minimum, the item name, National Stock Number, quantity, contract number and appropriate address markings on each unpacked item or shipping container. The marking shall be applied with a water-

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proof material and shall provide a definite contrast.

6. NOTES

6.1 Intended use. Trucks described herein are intended primarily for loading and unloading palletized cargo from military aircraft and will also be used for stacking, unstacking, and moving cargo in and around warehouses, loading platforms and docks within the military supply system. Trucks are intended for operation over paved, semiprepared and other hard surfaces, carrying the rated load for both short distances and extended travel as far as four miles of continuous operation.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Specification sheet title, number, and date.
(c) Time frame required for submission of the first-produced truck(s) and number of trucks required (see 3.2).
(d) When the Government will conduct any or all of the first-produced truck examination and tests. When the Government will conduct some but not all of the first article truck examination and demonstrations, the contracting officer should specify which examination and demonstrations will be conducted by the Government and which examination and demonstrations shall be conducted by the contractor (see 3.2).
(e) When a type DS truck is required (see 3.4).
(f) When nuclear safety is required (see 3.4.1, 3.4.2, 3.8, 3.9, 3.11.2, 3.22.2, and 3.22.6).
(g) When a rear view mirror is required (see 3.4.3).
(h) When a transmission oil sampling valve shall be furnished (see 3.6.1.7).
(i) Omitted.
(j) When a personnel cab is required (see 3.21).
(k) When final color shall be other than forest green (see 3.23).
(l) When other than operational use is required, specify service and adjustment required (see 4.5.3).
(m) Time frame for submitting first article demonstration report (see 4.6).
(n) Level of preservation, packaging, and packing required (see 5.1).
(o) Winterization when required (see 3.26).

6.3 First article truck. Any changes or deviations of production trucks from the approved first article truck during production will be subject to the approval of the contracting officer. Approval of the first article truck will not relieve the contractor of his obligation to furnish trucks conforming to this

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specification.

6.3.1 Incident report. A written report shall be furnished within 24 hours of any incident of equipment malfunction or failure during the conduct of a demonstration. As a minimum, the report shall describe components and parts affected, demonstration and operating conditions, date of incident, hour meter reading, how detected, description of incident, and corrective actions taken.

6.4 Fail-safe hydraulic lift system. A system will inherent safety features which will automatically prevent load lowering and mast tilting in the event of hydraulic line failure and provide a means of manually overriding the system to safely and slowly lower the load or empty carriage so that repairs can be made without incident.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each truck. When nuclear safety is required, manuals shall include those items required herein for nuclear safety. When a fail-safe hydraulic system is provided, manuals shall include a method for system verification by the operator and troubleshooting/repair procedures by maintenance personnel.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of trucks.

6.7 Rated capacity load. Rated capacity load shall be established with a load equivalent to an unrestrained cube weighing 10,000 pounds with overall dimensions twice the load center dimensions and whose center of gravity is located at the geometrical center of the cube. The actual load, with tolerances, is defined in 4.4.2.

6.8 Load center. both the horizontal and vertical load centers shall be 48 inches and shall be defined as follows:

a. The horizontal longitudinal distance from the intersection of the horizontal load carrying surfaces and the vertical load engaging faces of the forks to the center of gravity of the load.

b. The vertical distance from the horizontal load carrying surfaces of the forks to the center of gravity of the load.

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PREPARING ACTIVITY:
AIR FORCE - 84

CUSTODIAN:
AIR FORCE - 99

PROJECT NUMBER:
3930-F007

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APPENDIX

FOR

TRUCKS, LIFT, FORK, DIESEL

10. SCOPE. The demonstration methods contained in this appendix are to determine whether trucks procured under this specification conform to the requirements set forth.

20. APPLICABLE DOCUMENTS. None.

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DEMONSTRATION METHOD NO. 1

MAXIMUM FORK HEIGHT - FREE LIFT HEIGHT

COLLAPSED MAST HEIGHT

Demonstration course:

Level flat surface:

Demonstration procedure:

(a) Measure the true vertical distance from floor to uppermost projection of the upright assembly with mast in true vertical position. (This is the actual collapsed mast height).

(b) Raise rated load in a horizontal position until the inner mast begins to exceed the specified collapsed mast height.

(c) Measure and record the true vertical distance to the top surface of the forks at the specified load center. (This is free lift height).

(d) Adjust rated load to 12 inches, plus or minus 4 inches, above ground level and position load in true horizontal position.

(e) Attach a plumb bob to the rated load at the load center at the bottom and mark a reference point on the ground.

(f) Elevate the demonstration load to the maximum height and check the reference point. If necessary, reposition the mast without moving the truck so that the plumb bob is directly over the original reference point.

(g) Measure and record the true vertical distance from floor to the top surface of the forks. (This is maximum fork height).

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DEMONSTRATION METHOD NO. 2

LIFTING SPEED - LOWERING SPEED

1. Demonstration procedures:

(a) Measure and record distance in inches from floor to top surface of forks in fully lowered position.

(b) Record time in seconds required to raise rated load to maximum fork height using timing device. Record the difference between maximum fork height and the height of the forks in the lowered position. Calculate lifting speed.

(c) Record time to lower rated load at maximum speed (full open valve) to 3 to 4 foot height. Load to be abruptly stopped at the 3 to 4 foot height. Measure and record vertical distance from ground to top surface of forks. Calculate loaded lowering speed.

(d) Repeat steps (b) and (c) a minimum of three times. Calculate the average speeds.

(e) When nuclear safety is required, determine lowering speed immedately before forks contact ground. If a single stage lift cylinder is provided, no additional demonstrating is required but lowering speed determined above shall not exceed 65 feet per minute. If a multi-stage lift cylinder is provided, forks with rated load shall be lowered until further lowering will cause constant speed descent. Record time to fully lower from this point and the distance traveled by forks. Calculated lowering speed shall not exceed 65 feet per minute.

(f) Record time in seconds required to lower forks with no load, at maximum speed, from maximum lift height to ground level. Calculate the unloaded lowering speed.

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DEMONSTRATION METHOD NO. 3

RIGHT ANGLE TURN

1. Demonstration course:

   (a) Level paved surface.

   (b) Demonstration course in accordance with course layout (see figure A-1).

2. Demonstration procedure:

   (a) Truck is placed with the most forward edge of centered rated load coincident to line "B" (see course layout). When forks are longer than rated load, the fork tips shall touch line "B" and the fork length is included in the "A" dimension. Mast in vertical position.

   (b) Pallet is raised 6 inches above floor.

   (c) Position movable vertical barrier within 4 inches of counterweight of truck.

   (d) Turn steer wheels to the extreme left position and back truck through a 90-degree turn.

   (e) Measure the perpendicular distance from line "B" to the position assumed by the movable barrier. This is dimension "A."

   (f) Return truck to initial position.

   (g) Turn steer wheels to extreme right and repeat (a) through (e).

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Test Method No. 3

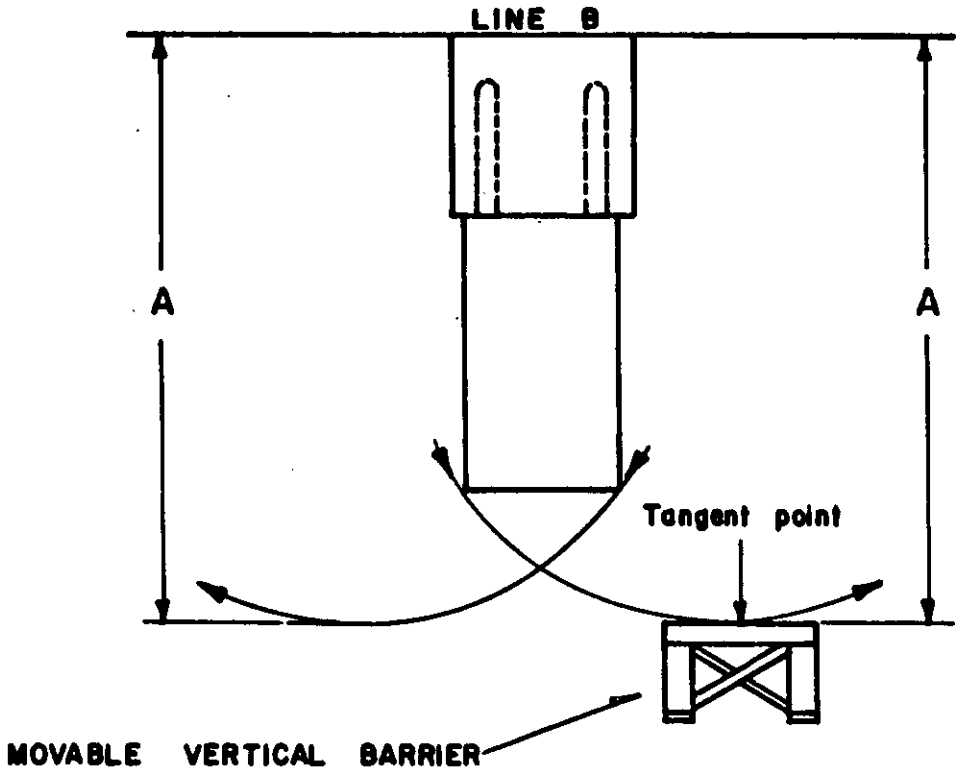


Figure A-1. COURSE LAYOUT

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DEMONSTRATION METHOD NO. 4

SPEED

1. Demonstration course:

Demonstration course shall be a dry, level surface of sufficient length to attain specified speed prior to entering measured distance, plus measured distance, plus sufficient stopping distance.

2. Demonstration procedure:

(a) Operate the truck at governed speed with rated load in carry position until all systems are at normal operating temperature.

(b) Drive the truck a sufficient distance to attain specified speed prior to entering the measured demonstration course.

(c) Record length of measured distance, and time to traverse measured distance. Measured distance shall be equal to or greater than 44 feet.

(d) Repeat for a total of six runs, three in highest forward gear and three in highest reverse gear, except only time measurement is required. Calculate the speed for each run and an average in each direction.

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DEMONSTRATION METHOD NO. 6

UPRIGHT TILT

1. Demonstration course:

Level, flat surface.

2. Demonstration procedures:

(a) Tilt.

(1) Place truck without load on flat surface.
(2) Raise forks to an elevation of approximately 2 feet above the surface, tilt boom as far forward as possible, and record angle of tilt.
(3) Tilt boom as far as possible and record angle of tilt.

(b) Tilt cavitation.

(1) Tie down rear of the truck to prevent overturning.
(2) Install pressure gage to read hydraulic pressure at head end of tilt cylinder.
(3) Raise rated load to maximum lift height with mast vertical.
(4) Without adjustment of height control, tilt load to full forward tilt at engine low idle by opening forward tilt valve completely (no featering of valve is permitted).
(5) Record lowest pressure reading at tilt cylinder.
(6) Repeat b(3), (4), (5) by feathering valve to produce slowest tilt speed practical.
(7) Record lowest pressure reading at tilt cylinder.

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DEMONSTRATION METHOD NO. 7

DRIFT

1. Demonstration course:

   Level, flat surface.

2. Demonstration procedure:

   (a) Heat the hydraulic oil by exercising the lift and tilt functions.
   (b) Raise rated load to maximum fork height obtainable with outer channel in true vertical position. Mark reference point on mast upper stage.
   (c) Record initial outer mast tilt.
   (d) Hold for specified time and record downward drift and rotational drift.

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DEMONSTRATION METHOD NO. 8

TRUCK OPERATIONS

Demonstration course:

(a) The demonstration course shall be a level, improved surface laid out in accordance with Figure A-2. Station 1 and Station 3 shall be platforms that are not less than 8 feet in height and capable of supporting rated load. Station 2 shall be at ground level.

(b) Obstacles shall be in accordance with Figure A-3 and placed so that the nearest obstacle is not less than 10 feet from Station 1.

Demonstration procedure:

(a) Initially, place rated loads on platforms at Station 1 and 3.

(b) Pick up load from Station 1; back up truck being sure all wheels travel over obstacles; travel forward and deposit load at Station 2.

(c) Disengage load at Station 2; back out; then travel forward to Station 3. Prior to engaging load at Station 3, raise forks to maximum lift height to activate relief valve. Lower forks and engage load.

(d) Pick up load at Station 3; back out; then travel forward and deposti load at Station 1, being sure all wheels travel over obstacles.

(e) Disengage load at Station 1; back out; being sure all wheels travel over obstacles; then travel forward to Station 2.

(f) Pick up load at Station 2; back out; then travel forward and deposit load at Station 3.

(g) Steps 2(b) through 2(f) shall constitute one circuit. For 50 percent of the circuits, Station 2 and Station 3 shall be interchanged so that they are in closkwise position rather then in their initial counterclockwise position. Stations shall be changed at the end of each 8-hour operating day.

(h) Truck with rated load shall be operated for a minimum of 40 hours.

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DEMONSTRATION METHOD NO. 8

TRUCK OPERATIONS

(i) Record total circuits for each hour of operation.

(j) Truck shall complete 6 to 8 circuits per hour.

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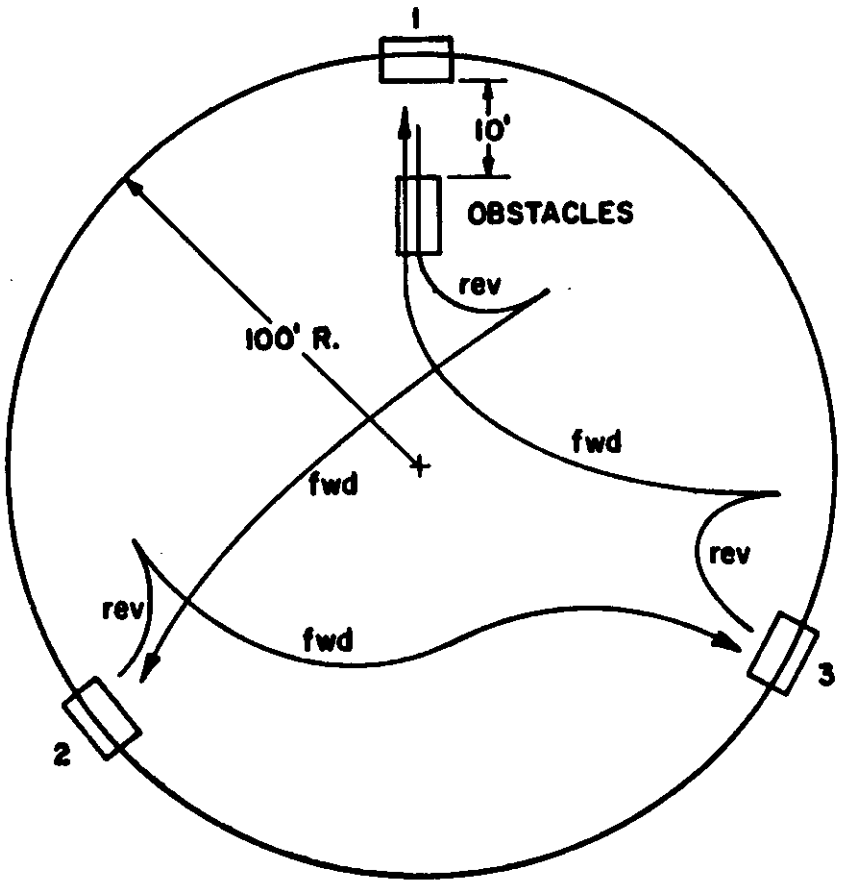
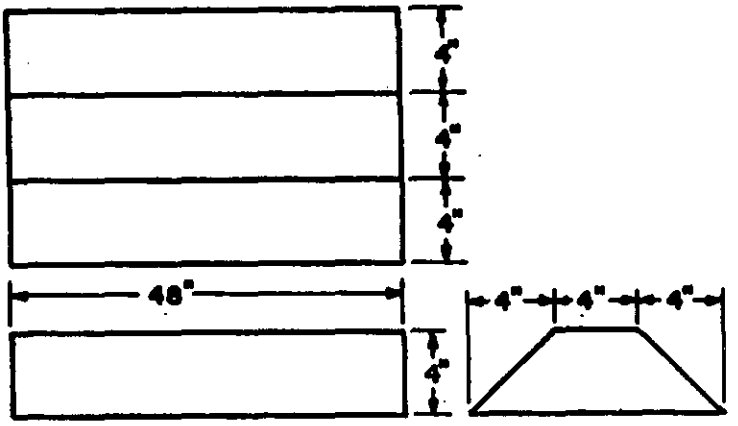


Figure A-2 TRUCK OPERATIONS COURSE

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LAYOUT OF OBSTACLE COURSE

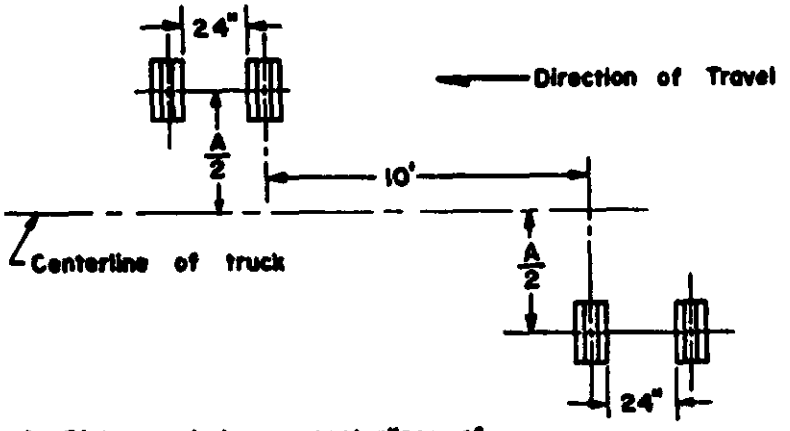


Figure A-3 DETAILS OF OBSTACLE CONSTRUCTION

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DEMONSTRATION METHOD NO. 9

CYLINDER PROOF

1. Test apparatus:

a. Hydraulic test bench or truck capable of developing 1.5 times system working pressure.

b. Pressure gauge.

c. Thermocouple.

2. Test procedure:

a. Double acting cylinder.

(1) Position and mechanically hold the piston at the approximate midpoint of the cylinder.

(2) Fill both sides of the piston with oil at not less than 120 degrees F.

(3) With the head end port capped, apply an oil pressure equal to 1.5 times the working pressure to the rod end of the piston for not less than 60 seconds. Release pressure.

(4) With the rod end port capped, apply an oil pressure equal to 1.5 times the working pressure to the head end of the piston for not less than 60 seconds. Release pressure.

b. Single acting cylinder.

(1) Extend cylinder to maximum extension by filling with oil at not less than 120 degrees F.

(2) Apply oil pressure equal to 1.5 times the working pressure at the inlet port for not less than 60 seconds. Release pressure.

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